

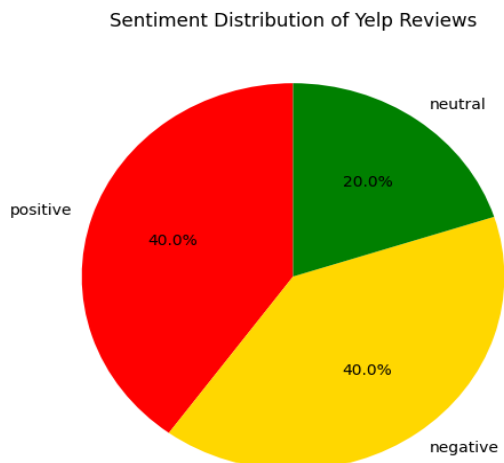
Q1: Analyze the basic statistics of your dataset

	label	text	sentiment	cuisine	price_range
0	4	dr. goldberg offers everything i look for in a...	positive	American	\$
1	1	Unfortunately, the frustration of being Dr. Go...	negative	Mexican	\$
2	3	Been going to Dr. Goldberg for over 10 years. ...	positive	Chinese	\$\$\$
3	3	Got a letter in the mail last week that said D...	positive	Japanese	\$\$\$\$
4	0	I don't know what Dr. Goldberg was like before...	negative	Japanese	\$\$

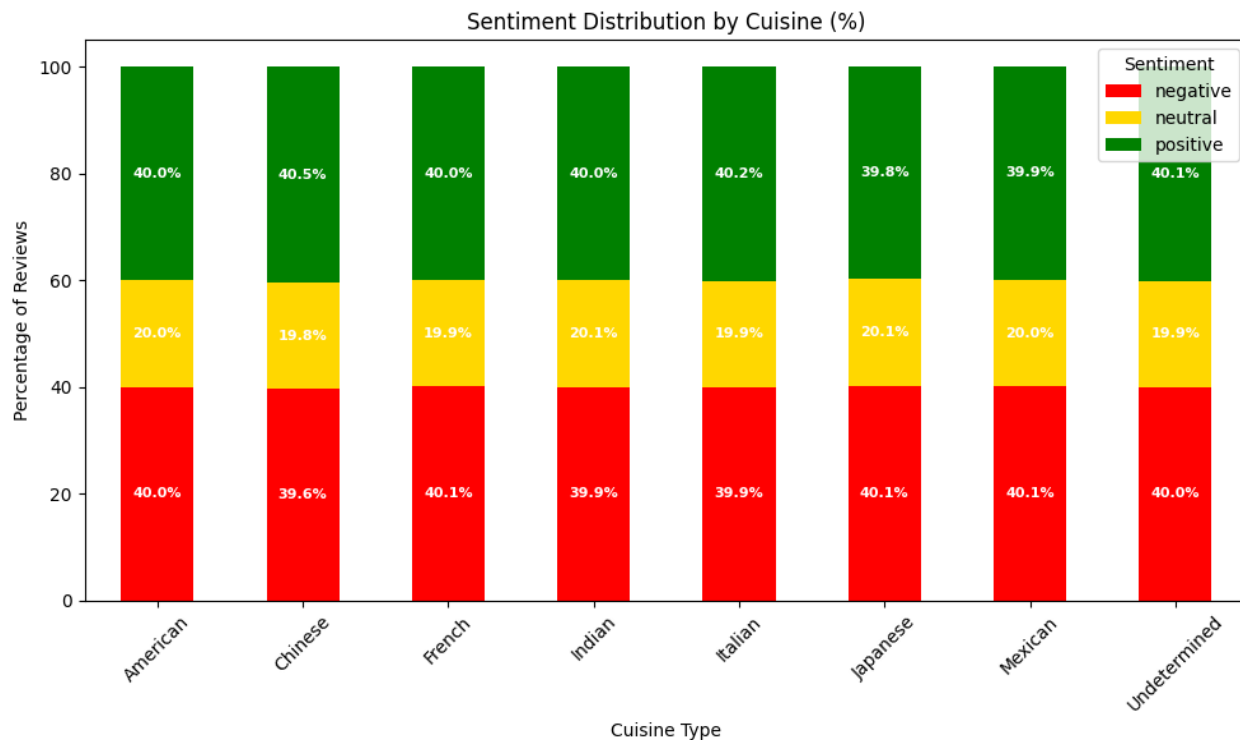
Q2: Analyze the data structure

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 650000 entries, 0 to 649999
Data columns (total 5 columns):
#   Column      Non-Null Count  Dtype
---  -
0   label       650000 non-null  int64
1   text        650000 non-null  object
2   sentiment   650000 non-null  object
3   cuisine     650000 non-null  object
4   price_range 650000 non-null  object
dtypes: int64(1), object(4)
memory usage: 24.8+ MB
```

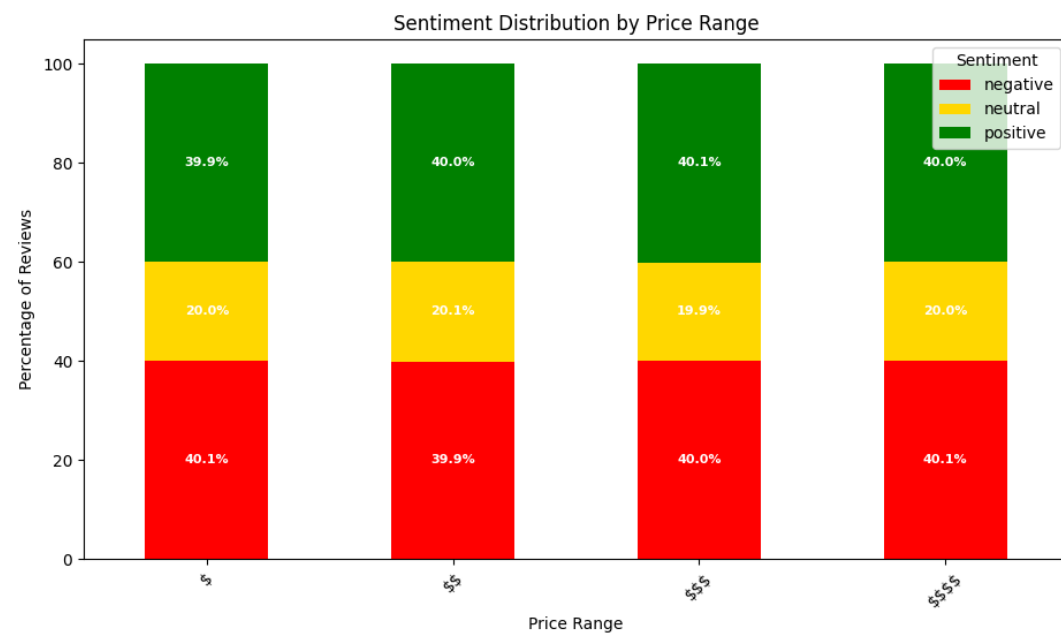
Q3: Create a pie chart of the sentiment distribution



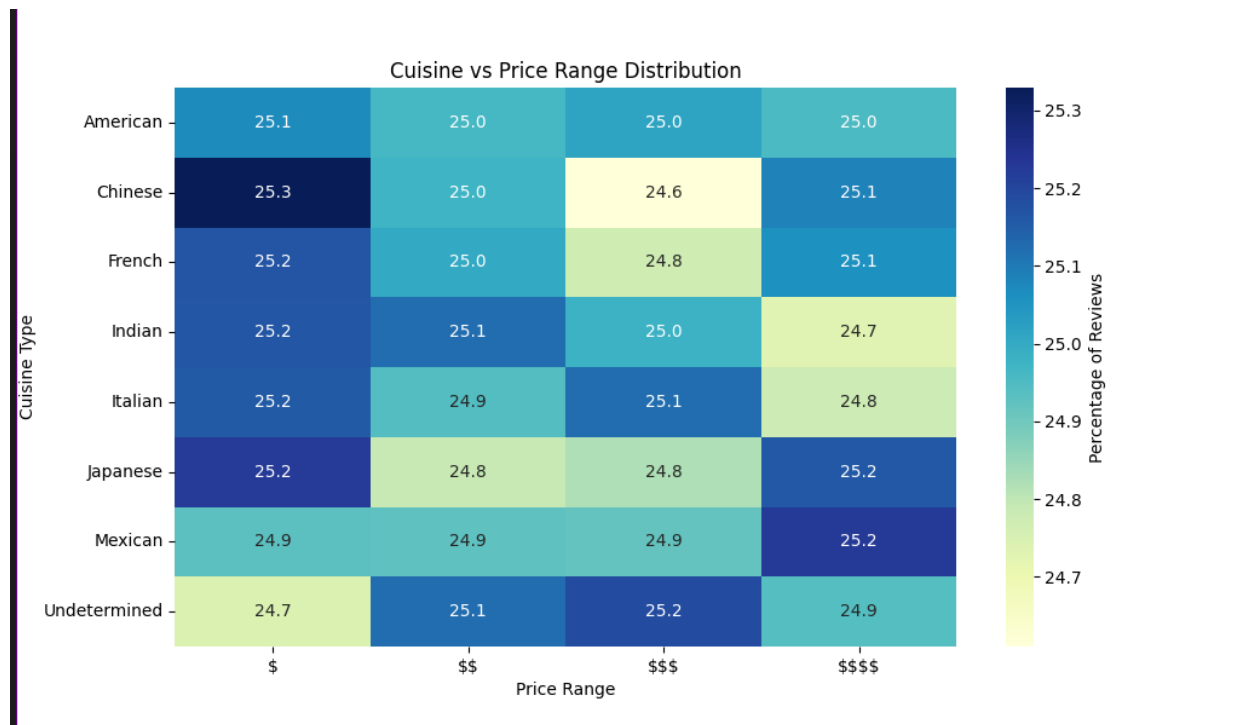
Q4: Analyze the sentiment patterns across different cuisines



Q5: Analyze the sentiment patterns across price ranges



Q7: Analyze the relationship between cuisine types and price ranges using a heatmap.



Q8: Summary

The overall dataset statistics were based on a total of approximately 650,000 reviews. The distribution across star ratings were balanced with approximately 130,00 per rating. The sentiment distribution was of roughly equal proportions in the context of positive and negative reviews with a slightly smaller neutral middle, which makes sense given Yelp's 1-5 star rating balance. The distributions of sentiment were also equal across the various cuisines, which is probably a direct result of their artificial assignments, yet the stack bar plots demonstrate a clear method of comparing customer satisfaction with different cuisine types. The price ranges were also distributed evenly, with the heatmap illustrating how different cuisines appear across all four pricing ranges.